MEDINOVA TEACHING HOSPITAL

Class 19 — Pharmacology & Prescribing

Safe Prescribing · Drug Interactions · Adverse Drug Reactions · Pharmacokinetics

> *"Every drug is a poison.*
> *The dose alone makes the difference."*

BOARD CERTIFICATION SERIES | MODULE 19 of 20

SECTION 01

Pharmacokinetics: ADME

Understanding the life cycle of a drug in the body determines your dosing intervals.

Key Formulas:
- **Volume of Distribution (Vd):** $Vd = \text{Dose} / \text{Plasma Concentration}$
- **Half-life ($t_{1/2}$):** $t_{1/2} = (0.693 \times Vd) / \text{Clearance}$
- **Steady State:** Achieved after 4-5 half-lives.
- **Loading Dose:** $LD = Vd \times \text{Desired Plasma Concentration}$

Phase	Key Concept	Clinical Implication
Absorption	Bioavailability (F)	IV has 100% bioavailability; Oral varies (First pass).
Distribution	Protein Binding	Low Albumin = More free (active) drug (e.g., Phenytoin).
Metabolism	CYP450 System	Interactions are common here (Inhibitors/Inducers).
Excretion	Clearance (Cl)	Determines maintenance dose (MD).

> **Prof. Obi's Pearl #1** "A loading dose gets you to the therapeutic window fast, but the maintenance dose keeps you there. If you give a loading dose without adjusting for renal clearance, you are just fast-tracking toxicity."

SECTION 02

Safe Prescribing & Renal Adjustment

The 10 Rights: Patient, Drug, Dose, Route, Time, Documentation, Reason, Response, Education, To Refuse.

eGFR Dose Adjustment (Common Examples)

Drug	eGFR 30–60	eGFR < 15
Metformin	Review dose (Max 1g)	STOP (Lactic Acidosis)
Enoxaparin	Standard dose	Reduce to 1mg/kg OD
Gentamicin	Extend interval	Avoid / Monitor levels strictly
Nitrofurantoin	Use with caution	STOP (Ineffective)

SECTION 03

High-Risk Medications

- **Insulin:** Never abbreviate "U". Use dedicated insulin syringes.
- **Opioids:** Calculate "Breakthrough" as 1/6th of the total daily dose.
- **Anticoagulants:** Cross-check INR before every Warfarin dose.
- **Concentrated K+:** NEVER IV PUSH. Must be diluted and infused slowly via pump.

SECTION 04 & 05

Drug Interactions & Adverse Reactions

CYP450 Metabolism

Inhibitors (Increase Drug Levels)	Inducers (Decrease Drug Levels)
Sodium Valproate | **I**soniazid	**P**henytoin | **C**arbamazepine
Cimetidine | **K**etoconazole	**R**ifampicin | **A**lcohol (Chronic)
Erythromycin | **G**rapefruit Juice	**P**henobarbital | **S**t. John's Wort

*Mnemonic: **SICKFACES.COM** (Inhibitors) | **PC BRAS** (Inducers).*

WHO ADR Classification

- **Type A (Augmented):** Dose-dependent, predictable (e.g., Bleeding with Warfarin).
- **Type B (Bizarre):** Not dose-dependent, idiosyncratic (e.g., Penicillin Anaphylaxis).
- **Type C (Chronic):** Long-term use (e.g., Osteoporosis with Steroids).
- **Type D (Delayed):** Teratogens or Carcinogenesis.

> Prof. Obi's Pearl #2 "The most common drug interaction in Lagos isn't on this list—it's between the hospital's prescription and the 'Herbal Supplement' the patient is taking in secret. Always ask about the green bottles."

SECTION 06

Prescribing in Special Populations

Elderly: The Beers Criteria / STOPP-START

Avoid in Elderly: Long-acting Benzodiazepines, Anticholinergics (Amitriptyline), and NSAIDs (High risk of GI bleed/AKI). **"Start Low, Go Slow."**

Pregnancy Drug Categories

Category	Risk Level	Action
A	Safe	No evidence of risk (e.g., Folic Acid).
B	Likely Safe	Animal studies show no risk.
C	Caution	Use only if benefit outweighs risk.
D	Evidence of Risk	Avoid unless life-threatening.
X	TERATOGENIC	Absolute contraindication (e.g., Isotretinoin).

> Prof. Obi's Pearl #3 "Treating a patient with liver failure is like driving a car without brakes. You can't rely on the body to stop the drug's effect. Use drugs that are renally excreted or short-acting."

SECTION 07

Controlled Drugs (Schedules)

Schedule	Requirements	Examples
1	No medicinal use. Research only.	LSD, Cannabis (some regions).
2	Locked cupboard, CD Register, 28-day validity.	Morphine, Fentanyl, Ketamine.
3	Stronger prescription requirements.	Tramadol, Midazolam, Pregabalin.
4	Part 1 (Benzos) / Part 2 (Steroids).	Diazepam, Testosterone.
5	Low strength / OTC.	Codeine Linctus.

Prescription Requirements for CD (Sch 2/3): Must state the Form, Strength, and **Total Quantity in Words and Figures** (e.g., 10 (TEN) tablets).

> Prof. Obi's Pearl #4 "A prescription is a legal document, not a suggestion. If you make a mistake on a controlled drug script, you don't just risk the patient—you risk your license. Double-check the words and the figures."

OFFICIAL SIGN-OFF

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INSTITUTION: MediNova Teaching Hospital
SERIES: Board Certification Module 19 of 20
DATE: March 2026